
ME 642: Computational modelling of granular flows

Course Code	ME 642
Course Name	Computational modelling of granular flows
L-T-P-C	3-0-2-4
Prerequisites	Continuum mechanics / fluid mechanics / solid mechanics / soil mechanics
Intended for	MTech / PhD
Mutual Exclusion	NA
Approval	60th BoA (Jun 2025)

Preamble

Granular materials are ubiquitous in nature and industry, spanning applications from landslides and avalanches to mineral processing and pharmaceutical manufacturing. Their complex flow behavior—exhibiting solid-like, fluid-like, and gas-like regimes—poses significant modeling challenges. This course, Computational Modelling of Granular Flows, offers a comprehensive foundation in theoretical, discrete, and continuum modeling approaches. Topics include particle-scale simulations using the Discrete Element Method (DEM), continuum formulations with depth-averaged models, and the Kinetic Theory of Granular Flows (KTGF). Coupled CFD–DEM techniques and emerging machine learning methods will also be covered. Students will gain hands-on experience with open-source software like LIGGGHTS, OpenFOAM, and CFDEM, enabling them to model and analyze granular flows in diverse engineering systems.

Course Modules

- **Introduction to granular flows and theoretical foundations:** Granular materials in nature and industry; particle properties: shape, size distribution, cohesion; flow regimes: quasi-static, dense, and dilute. Experimental characterization of granular flows; stress, strain, and constitutive models; flow instabilities, segregation; governing equations for particulate flows. (6 lectures)
- **Discrete element method:** DEM theory and contact models; normal and tangential forces, friction; time integration schemes, stability considerations; particle-wall interaction and boundary conditions; interaction with STL files; calibration and validation, hands-on with open-source LIGGGHTS code; applications: mixing and segregation, hoppers, chutes, vibrating screens. Principles of CFD-DEM coupling; inter-phase forces, multiphase flow applications, case studies in mineral processing and energy systems; use of open-source CFDEM code. (17 lectures)
- **Continuum modelling: depth-averaged model and rheologies:** Introduction to continuum modeling of granular flows; derivation of depth-averaged / shallow water mass and momentum equations; basal friction laws, source terms, Savage-Hutter model; numerical solution strategies for hyperbolic PDEs; analytical and numerical solutions and analyses; applications to landslides, avalanches and debris flows; demonstration of open-source codes and integration with digital elevation models; application of complex rheologies to dry and wet granular flows – including fluid-like (Newtonian, non-Newtonian), plastic (Mohr-Coulomb, Drucker-Prager) and intermediate (such as μ -I rheology). (9 lectures)

- **Kinetic theory of granular flows (KTGF):** Fundamentals of KTGF and application to dense granular flows; collisional dissipation (inelastic interactions); granular temperature equation; granular pressure; derivation of constitutive relations and types of granular viscosities; Eulerian two-fluid models using KTGF; applications in fluidized beds; hands-on with software (ANSYS Fluent, OpenFOAM) (6 lectures)
- **Advanced topics:** Handling non-spherical particles; cohesion; wet granular flows; machine learning applications in granular flows. (4 lectures)

Laboratory Modules

- **DEM:** Discrete element modelling in LIGGGHTS open-source code. Solve problems on mixing and segregation, hoppers, chutes, vibrating screens, and granular column collapse. (8 hours)
- **Depth-averaged modelling:** Simulation of geotechnical flows (landslides and snow avalanches) using depth-averaged continuum method in open-source and commercial codes. (6 hours)
- **KTGF:** Multiphase Eulerian modelling of pneumatic conveying in ANSYS Fluent. (4 hours)
- **CFD-DEM:** Modelling and simulation of gas-solid fluidisation and mixing in a vertical fluidised bed using open-source and commercial codes. (6 hours)
- **Machine learning:** Supervised ML using Scikit-learn, dataset preparation (using previous lab results), regression and classification. (4 hours)

Text Books

1. Rao, K. Kesava, Prabhu R. Nott, and S. Sundaresan. *An introduction to granular flow*. Vol. 490. Cambridge: Cambridge university press, 2008.

References

1. O'Sullivan, Catherine. *Particulate discrete element modelling: a geomechanics perspective*. CRC Press, 2011.
2. Thornton, Colin. *Granular Dynamics, Contact Mechanics and Particle System Simulations: A DEM study*. Springer, 2016.
3. Pudasaini, Shiva P., and Kolumban Hutter. *Avalanche dynamics: dynamics of rapid flows of dense granular avalanches*. Springer Science & Business Media, 2007.
4. Gidaspow, Dimitri. *Multiphase flow and fluidization: continuum and kinetic theory descriptions*. Academic press, 1994.
5. Norouzi, Hamid Reza, et al. *Coupled CFD-DEM modeling: formulation, implementation and application to multiphase flows*. John Wiley & Sons, 2016.